

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GPH 423 — Past Questions & Materials
by Renance

Course: GPH 423
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — GPH 423 PQ 2024

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



DEPARTMENT OF EARTH SCIENCES

SECOND SEMESTER: B. SC. (GEOPHYSICS) 2023/2024 SESSION EXAMINATION

COURSE: GPH 423 (2 Units) RADIOMETRIC METHODS TIME ALLOWED: 2 Hrs.

INSTRUCTION: ANSWER QUESTION (1) AND ANY OTHER TWO (2) QUESTIONS

✓ **Question 1**
The given table shows the reading obtained with a gamma-ray spectrometer along a traverse perpendicular to foliation across a granite-gneiss outcrop near Igarra, Edo State, Nigeria.

Station (ft)	Spectrometer Reading (cpm)		
	T _c	U _c	K _c
0	13	28	195
100	8	27	243
200	25	36	218
300	15	27	193
400	15	30	197
500	8	21	233

Given that $k_1 = 0.6$, $k_2 = 0.13$, $k_3 = 0.02$, $S_1 = 1.0$, $S_2 = 1.5$, and $S_3 = 1.7$:

- Determine the Thorium (Th), Uranium (U), and Potassium (K) content at each station. (20 marks)
- Plot profiles for each element as well as a profile of the Th:U ratio (10 marks)

Question 2

- From the following transition of radionuclides, complete the equation and identify the particle emitted in the radioactive decays? (2 mark)
 - $^{235}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ??$ (2 mark)
 - $^{74}_{33}\text{As} \rightarrow ^{74}_{34}\text{Se} + ??$ (2 marks)
 - $^{235}_{92}\text{U} \rightarrow ^{207}_{82}\text{Pb} + ??$ (2 marks)
 - $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + ??$
- What do you understand by the following as used in radiometric prospecting? (4 marks)
 - Secular Equilibrium (4 marks)
 - Transient Equilibrium (4 marks)
 - No-Equilibrium

✓ **Question 3**

- Differentiate between half-life and mean-life (4 marks)
- Show that the half-life of an element is $T_{1/2} = 0.693\lambda^{-1}$ (4 marks)
- How long will it take for 18.0 grams of Ra-226 to decay to leave a total of 2.25 grams? (6 marks)
- Ra-226 has a half-life of 1600 years
- If the half-life of 100.0 grams of a radioactive isotope is 8 years, how many grams will remain in 32 years? (6 marks)

Question 4

- Mention THREE (3) principal devices used for detecting radioactive radiations, stating the type of radiation they detect/respond to and mode of surveying (ground base, airborne or both) (8 marks)

$k_3(k_c - S_3(u_c - S_1 T_c)) - S_3 T_c$

b) Discuss the various modes of interaction of gamma rays with matter.

(12 marks)

Question 5

a). What do you understand by the term as it relates to radioactive decay:

(i) Activity

(ii) Specific Activity

(2 marks)

(2 marks)

b). The isotope of ^{57}Co decays by electron capture to ^{57}Fe with a half-life of 272 days. ^{57}Fe nucleus is produced in an excited state, and it almost instantaneously emits gamma rays that can be detected.

(i) Find the decay constant of a ^{57}Co radiation

(6 marks)

(ii) If the activity of a ^{57}Co radiation source is now $2.00\mu\text{Ci}$, how many ^{57}Co nuclei does the source contain.

(10 marks)

Not
R

Ther's principle